



Research Paper

Thermal management architecture optimization for single-phase immersion liquid cooling

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ABSTRACT

To enhance the power usage efficiency of data centers, the current work focuses on examining how the coolant properties and liquid cooling structures affect the performance of single-phase immersion liquid cooling (ILC) systems. The three-dimensional mathematical model of the flow and heat transfer in single-phase ILC tanks is constructed. It explores the impacts of the tank's import/export layout, the immersion depth of the simulated motherboard, and the heat sink's geometric structure on ILC performance. Moreover, it clarifies the internal relationship between coolant properties and the tank's cooling performance under heat sink structure changes. The study concludes that the import/export layout and immersion depth have minor effects on cooling performance, while the heat sink's geometric structure significantly influences ILC performance. The performance of ILC tanks reaches its optimum when the fin pitch is 2 mm and the fin thickness is 2 mm. Although the cooling performance of the ILC tank continuously improves as the fin height increases, this improvement will no longer be significant when the fin height exceeds 40 mm. The effect of the coolant properties on the ILC performance varies among different liquid cooling structures. Among them, the dynamic viscosity of the coolant has the most significant effect on the ILC performance. Furthermore, the degree of correlation between the coolant properties and the ILC performance is more sensitive to the fin pitch than to the fin height and fin thickness.

1. Introduction

As information technologies like artificial intelligence, internet of Things, and big data are advancing at a breakneck pace, the size and power density of data centers have been witnessing exponential growth. It is estimated that by 2030, data centers are expected to consume 10% of the total global electricity, and the thermal management systems alone will account for 30%–45% of their total energy consumption [1,2]. Against this backdrop, traditional air cooling technologies are insufficient to meet the thermal management demands of high-power servers [3]. In view of this, developing more efficient and energy-saving thermal management solutions has become a key technical challenge in the development of the new generation of data centers [4,5].

To effectively address the above-mentioned challenges, a series of emerging technologies have emerged successively, including liquid cooling [6–8], phase-change material thermal buffering [9,10], and heat pipe cooling [11,12]. Among these technologies, single-phase

immersion liquid cooling (ILC) has achieved extensive attention because of its excellent heat transfer capability, compact structural design, and lower energy consumption level [13,14]. Numerous scholars have fully confirmed the remarkable superiority of the single-phase ILC technology through theoretical analysis and experimental research. For example, Ge et al. [15] used numerical simulations to conduct a comparative analysis of the performance of data centers using single-phase ILC and air cooling conditions. The research results show that the ILC system has obvious advantages compared with the traditional air cooling method under the same Reynolds number. Its maximum temperature is significantly reduced by 70.1%, the convective heat transfer coefficient is increased by 2.9 times, and it performs better in terms of heat transport uniformity. Choi et al. [16] constructed a new type of single-phase ILC systems. Compared with the traditional liquid cooling plate, the peak temperature and the maximal temperature variation of electronic devices in this new single-phase ILC system are decreased by 6.7 °C and 3 °C, respectively, effectively improving the cooling effect.

To further enhance the performance of single-phase ILC systems and better meet the stringent thermal management requirements of future

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Nomenclature			
A	surface area of heat sinks	Q_v	volume flow rate
c_p	specific heat capacity	x, y, z	coordinate axes
$C_\mu, C_{1\varepsilon}, C_{2\varepsilon}$	model parameters	<i>Greek symbols</i>	
D	fin pitch	ΔP	pressure drop
G_k	generation of turbulent kinetic energy	ΔT	average temperature variation
H_f	fin height	ε	turbulent dissipation rate
H_i	immersion depth	λ	thermal conductivity
w	fin thickness	μ	dynamic viscosity
k	turbulent kinetic energy	μ_t	turbulent viscosity
l	characteristic length	ρ	density
N	number of grid cells	$\sigma_k, \sigma_\varepsilon$	model parameters
$\overline{Nu}$	average Nusselt number	<i>Abbreviations</i>	
p	pressure	3D	three-dimensional
$\bar{p}$	average pressure	CPU	central processing unit
P	chip power	DIMM	dual inline memory module
T	temperature	HDD	hard disk drive
$\bar{T}$	average temperature	ILC	immersion liquid cooling
T'	pulsation temperature	MB	motherboard
T_{max}	maximum temperature	PCB	printed circuit board
V	velocity	PCH	platform controller hub
$\bar{u}$	average velocity	PSU	power supply unit
u'	pulsation velocity		

high-power servers, current research on single-phase ILC mainly focuses on aspects such as coolant selection [17–20], liquid-cooling structure optimization [21,22], operating condition regulation [23–25], and heat transfer enhancement technologies [26–28]. In terms of coolant selection, Jithin and Rajesh [13] conducted in-depth numerical study to analyze the impact of three different coolants on the single-phase ILC performance. It was found that under the premise of achieving the same cooling effect, the energy consumption of the single-phase ILC system from high to low is as follows: mineral oil, electronic fluorinated liquid, and ultrapure water. Wang et al. [20] also studied the influence of different cooling oils on the immersion cooling performance through numerical simulation. The results show that compared with mineral oil and synthetic oil, the Nusselt number of silicone oil with high viscosity is reduced by 24.4%, and the flow resistance loss is significantly increased by 7.4 times. The design of the liquid-cooling structure plays a crucial role in the flow and heat transfer characteristics inside the single-phase ILC system. Shrigondekar et al. [21] optimized the liquid-cooling structure of the single-phase ILC system through experimental research. The research reveals that compared to a heat sink featuring a larger fin pitch, the thermal resistance of that with a smaller fin pitch is elevated by 19% to 43%. Moreover, compared with U-shaped and Z-shaped layouts, the temperature and thermal resistance of the single-phase ILC system with a T-shaped inlet and outlet layout are reduced by up to 4.9% and 12.6%, respectively. Huang et al. [22] numerically simulated the gravity-driven single-phase ILC system and the buoyancy-driven single-phase ILC system. The results show that the PUE of the pump-driven single-phase ILC system is 11.6% lower than that of the gravity-driven one and exhibits better heat transfer uniformity. In practical engineering applications, the operating conditions of the single-phase ILC system vary greatly. Therefore, a deep understanding of the control mechanism of operating conditions is of great value for the effective management of the single-phase ILC system. Cheng et al. [23] studied the performance of the pump-driven single-phase ILC system of workstations under different flow rates. The findings indicated that an increase in the coolant flow rate from 0.4 m/s to 1.2 m/s led to a 64.3% reduction in the temperature difference index of the device temperature. Liu et al. [24] researched the cooling performance of a lithium-battery single-phase ILC system. They found that as the discharge rate increases,

the lithium-battery temperature shows an upward trend. Huang et al. [25] experimentally studied the pump-driven single-phase ILC system under different working conditions. They found that there is a critical flow rate. When this critical value is exceeded, further increasing the coolant flow rate has no obvious improvement effect on the single-phase ILC performance. To further improve the performance of the single-phase ILC system, scholars have also developed some active and passive heat transfer enhancement technologies. Muneeshwaran et al. [27] introduced heat-pipe technology into the single-phase ILC system to design a novel heat sink with heat pipes. The study shows that under the same conditions, the novel heat sink can further reduce the temperature of electronic devices by 4.8 °C compared with the conventional one. Ding et al. [28] proposed an active heat transfer enhancement technology for the single-phase ILC system that couples fans and micro-pumps for high-heat-flux servers, which further reduces the temperature of electronic devices by 5%–6.1%.

Through the investigation of research related to single-phase ILC, it can be found that existing studies mainly focus on the impact of liquid cooling structure and coolant thermophysical properties on the single-phase ILC performance and the energy efficiency improvement technologies of single-phase ILC systems. However, most of the relevant studies on the influence of liquid cooling structure and coolant thermophysical properties only analyze the change trends, without deeply analyzing the internal influence mechanisms behind these trends. In particular, there is a lack of quantitative evaluation of the influence degree of different liquid cooling structure parameters on the single-phase ILC performance. Additionally, most of the existing literature studies the liquid cooling structure or coolant thermophysical properties separately, and few studies comprehensively analyze the coupling effect of the liquid cooling structure and coolant thermophysical properties. Moreover, the correlation mechanism by which the impact of coolant thermophysical properties on single-phase ILC performance is affected by liquid cooling structure parameters still needs to be further revealed.

To address these challenges, this study aims to deeply investigate the influence mechanisms of the coolant thermophysical property and the liquid cooling structure on the cooling performance of single-phase ILC systems. By establishing a three-dimensional (3D) steady-state model of the flow and heat transfer processes within a single-phase ILC tank, the

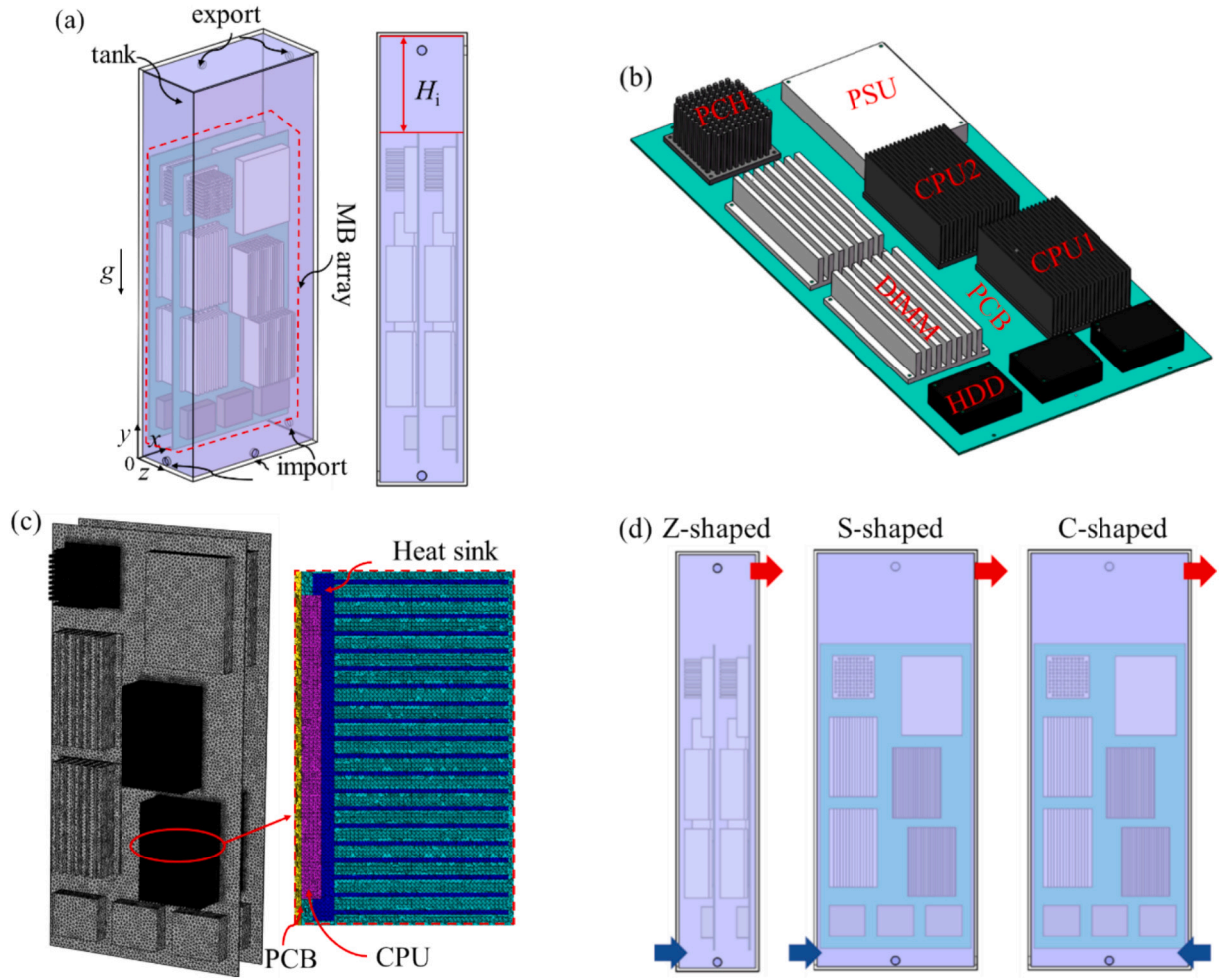


Fig. 1. Schematic diagram of the single-phase ILC tank: (a) 3D geometry, (b) MB layout (c) Grid system, (d) import/export layout. (CPU: central processing unit, PSU: power supply unit, DIMM: dual inline memory module, PCH: platform controller hub, HDD: hard disk drive).

Table 1
Information about key components.

Components	Dimensions (mm)	Heat sink	Power (W)	Heat flux (W/cm ²)	Materials
CPU	75 × 70 × 4.8	Base: 115 × 80 × 5 mm Fin: 115 × 1 × 40 mm Fin pitch: 3 mm Fin number: 20	350	6.67	stainless steels
PCH	50 × 50 × 4.8	Base: 70 × 70 × 5 mm Fin: 3 × 3 × 40 mm Fin pitch: 3 mm Fin number: 107	175	7	
PCB	500 × 250 × 2		0		FR4
DIMM	130 × 80 × 30	/	0		
PSU	130 × 100 × 20		0		
HDD	50 × 60 × 20		0		

effects of the import/export layout of the single-phase ILC tank and the immersion depth of the simulated motherboard (MB) on the single-phase ILC performance are explored. Furthermore, optimization research on the structure of the immersion heat sink is conducted to clarify the intrinsic relationship between the coolant properties and the flow and thermal transport characteristics of the single-phase ILC system, thereby providing theoretical guidance for practical engineering applications.

2. Numerical methodology

2.1. Physical model

To explore the influence of the liquid - cooling structure on the cooling performance in a single-phase ILC system, this paper constructs a simplified ILC tank for servers in data centers. As shown in Fig. 1(a), the ILC tank consists of a box and two simulated MBs, with the outer dimensions of 270 mm × 136 mm × 832 mm. The box is crafted from stainless steel having a thickness of 5 mm. Fig. 1(b) shows a simulated MB designed by imitating the actual blade server, which includes two CPUs and one PCH as heat - source modules, as well as a PSU, DIMMs, and an HDD as flow - resistance modules. The key information of each component on the simulated MB is given in Table 1. In this paper, the heat - source modules are made by pasting heating films on the bottom of stainless - steel blocks, and the heat sinks in close contact with the upper parts of the stainless - steel blocks are all made of 6061 aluminum

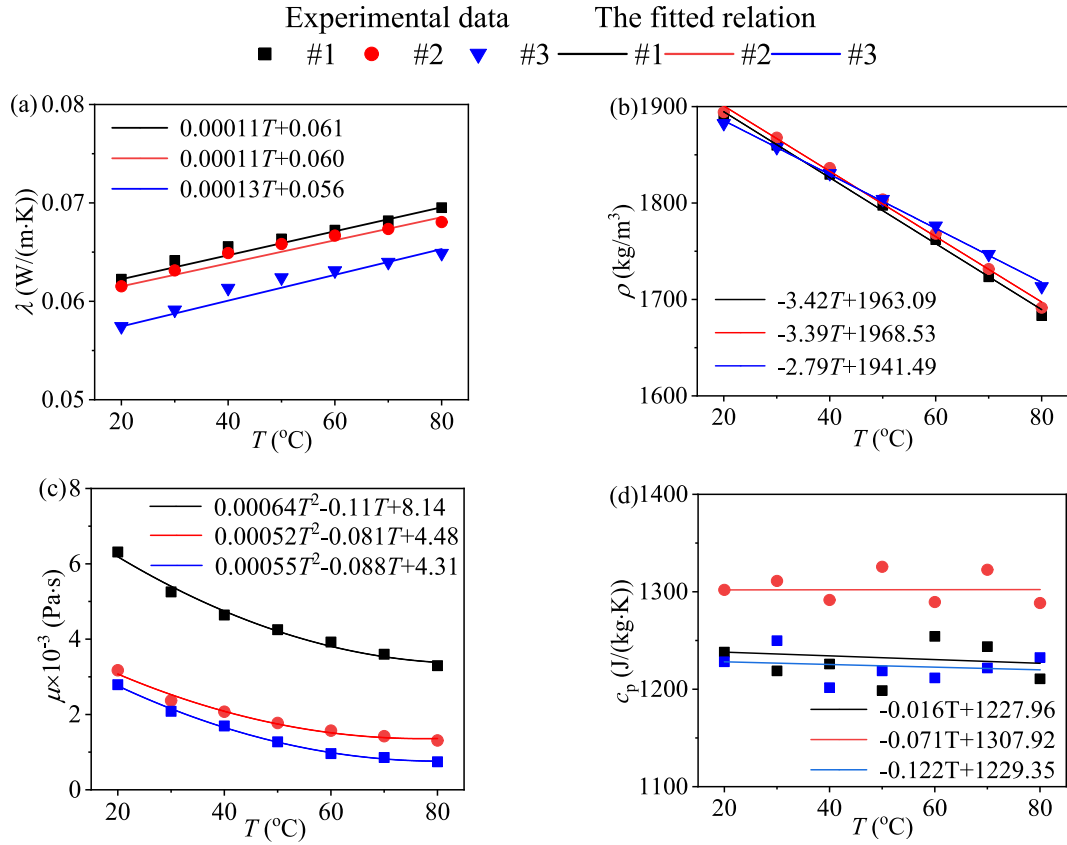


Fig. 2. Thermophysical properties of three coolants: (a) Thermal conductivity, (b) Density, (c) Dynamic viscosity, (d) Specific heat capacity.

alloy. It should be noted that, as thermal contact resistance only affects the absolute values of parameters such as temperature and thermal resistance, and does not alter the analysis of immersion cooling characteristics or their variation patterns, it is not considered in this study in order to simplify the calculations [29,30]. Fig. 1(c) presents the grid diagram of the computational domain in this paper. Most of the grids are unstructured grids composed of hexahedrons, and boundary - layer encryption is adopted near the heat sources and their heatsinks to improve the calculation accuracy. Notably, this paper defines the immersion depth (i.e., H_i) of the simulated MB, specifically the distance between the coolant liquid level and the top of the simulated MB, as shown in Fig. 1(a). The ILC characteristics at immersion depths of 50 mm, 100 mm, and 150 mm are investigated separately. Moreover, three different coolant flow patterns with Z - shaped, S - shaped, and C - shaped import/export layouts are designed, as shown in Fig. 1(d). In addition, three fluorinated fluids from Zhejiang Juhua Co., Ltd. are selected as the coolants. For the sake of simplified description, they are named as Coolant #1, Coolant #2, and Coolant #3 in this paper. Fig. 2 illustrates how the thermophysical properties of the coolant change with temperature. It should be noted that although the specific heat capacity varies nonlinearly with temperature, the deviation between the fitted curve and actual values is only 0.1%–2.5%. Therefore, a simplified treatment is adopted in this study. Moreover, all coolants have boiling points exceeding 100 °C, making them suitable for single-phase ILC applications.

2.2. Governing equations

In this study, a 3D steady - state model for the internal flow and heat transfer within a single - phase ILC tank is developed, relying on the previously established physical model. Here, the coolant is regarded as an incompressible fluid. Given the intricate nature of the liquid cooling

structure, the flow is presumed to be in a turbulent state. For the sake of simplicity, the wall thickness of the single - phase ILC tank and any heat loss to the surrounding environment are not taken into account.

Based on these assumptions, the Reynolds-averaged Navier-Stokes equations can be obtained.

$$\frac{\partial(\bar{u}_i)}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial(\bar{u}_i \bar{u}_j)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 \bar{u}_i}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_j} (\bar{u}_i' u_j') \quad (2)$$

$$\rho c_p \frac{\partial(\bar{u}_i \bar{T})}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} - \rho c_p \bar{u}_i' \bar{T}' \right) \quad (3)$$

where $\bar{u}_i$, $\bar{u}_j$ represents the average velocity in different axes, $\bar{T}$ denotes average temperature, $\bar{p}$ denotes average pressure, u_i' , u_j' and T' represent the pulsation values.

Given that the standard k- ϵ model is capable of simulating fully developed turbulent flow with high accuracy and reliability, and that the focus of this study is not on capturing the laminar-to-turbulent transition region—an area where the SST k- ω model holds a distinct advantage—and given that this model has been widely employed in the existing literature [29,31–33] to describe turbulent phenomena, it was therefore selected for the numerical simulations in this work.

$$\rho \frac{\partial(\bar{k} \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_k} + \frac{\partial k}{\partial x_j} \right) \right) + G_k - \rho \epsilon \quad (4)$$

$$\rho \frac{\partial(\bar{k} \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_k} + \frac{\partial k}{\partial x_j} \right) \right) + C_{1\epsilon} \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (5)$$

where k and ϵ represent the turbulent kinetic energy and energy

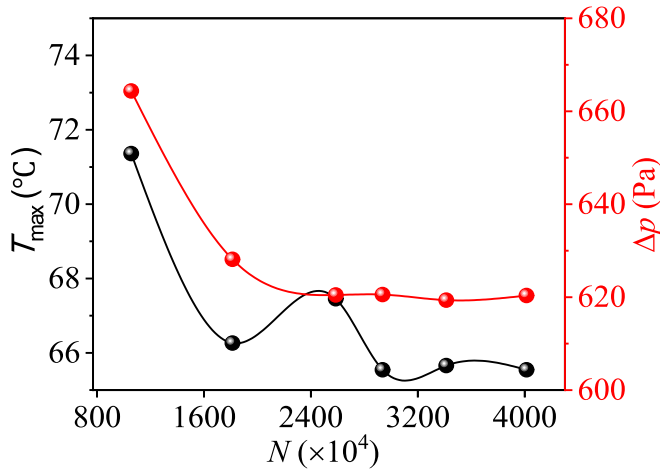


Fig. 3. Independence verification testing.

dissipation rate, G_k indicates the turbulent kinetic energy resulting from the average velocity gradient, and μ_t is turbulent viscosity.

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (6)$$

where C_μ is the correction parameter (i.e., $C_\mu = 0.09$). Other parameters are: $\sigma_k = 1.0$; $\sigma_\varepsilon = 1.3$; $C_{1\varepsilon} = 1.44$; $C_{2\varepsilon} = 1.92$.

In this paper, the inlet temperature and flow velocity of the coolant are 15°C and 0.65 m/s , respectively. The outlet type of the single-phase ILC tank is “outflow”. The tank walls are set as adiabatic boundary conditions, while the fluid – solid interface adopts the no - slip boundary and conjugate heat transfer boundary conditions.

2.3. Solution methods

This paper conducts calculations on the flow and heat transfer performance of a single-phase ILC system based on the ANSYS Fluent computing platform. During the solution process, the solver used is a double-precision pressure-velocity coupling solver, and the solution method is the finite volume method. During the iterative solution procedure, the momentum and energy equations are discretized using the second - order upwind scheme. The SIMPLE algorithm is used for velocity-pressure coupling, and PRESTO! is used for pressure iteration. In order to strike a balance between computational time and precision, the under - relaxation factors assigned to pressure, momentum, and energy within the calculation process are respectively set at 0.2, 0.8, and 0.9. Concerning the convergence control parameters, the residual value for the energy equation is fixed at 10^{-6} , and the residuals of the

remaining equations are set at 10^{-4} . Additionally, the iterative calculation stops when the net power consumption of the single-phase ILC system is 0 W and the outlet temperature and pressure are stable.

To eliminate the impact of the grid number on the simulation accuracy, this paper selects six different grid systems for an independence test. The coolant in the single-phase ILC tank is Coolant #1, and the immersion depth of the coolant is 100 mm . Fig. 3 reveals that once the grid number surpasses 3.2×10^7 , any additional increase in the grid number has negligible impact on both the maximum temperature rise of the CPU and the pressure difference between the inlet and outlet derived from numerical calculations. Evidently, when $N \geq 3.2 \times 10^7$, the effect of the grid number on the simulation outcomes can be largely disregarded. Notably, grid independence tests are not performed for every case in this study, as doing so would have been computationally prohibitive. To ensure solution accuracy, this study maintained a mesh density in the core heat source region, its adjacent boundary layers, and the fin gaps that was consistent with that of the validated cases—a common practice in numerical studies.

2.4. Model validation

To demonstrate the reliability of the numerical model and calculation results, this paper replicates the single-phase ILC experiment conducted by Huang et al. [25] for data center servers. Fig. 4(a) presents the schematic of the performance experiment of the single-phase ILC system. Notably, the size of the computational domain in the numerical model of this paper is consistent with the size of the ILC tank designed by Huang et al. [25], and the dimensions and materials of the corresponding electronic devices and heat sinks are also consistent with the experiment [25]. Moreover, the boundary conditions are consistent with the experimental operating conditions, i.e., the total power of the electronic devices is 1.75 kW , the temperature of the cooling water is 15°C , and the volume flow rate varies between 2 and 8 L/min . Fig. 4(b) plots the comparison of the CPU junction temperatures between the experiment and the numerical simulation. As illustrated in the figure, the CPU temperature obtained based on the numerical calculation is in high agreement with the experimental test results, and the maximum deviation between the two is 3.0% . This deviation is almost negligible in the scope of scientific research. Given this, it can be seen that it is reasonable and reliable to study the influence of different liquid cooling structures on the performance of the single-phase ILC system based on the simulation method and physical model in this paper.

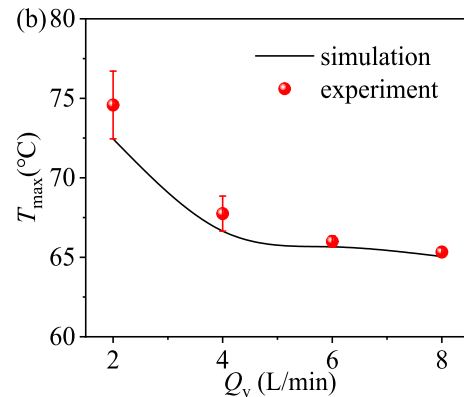
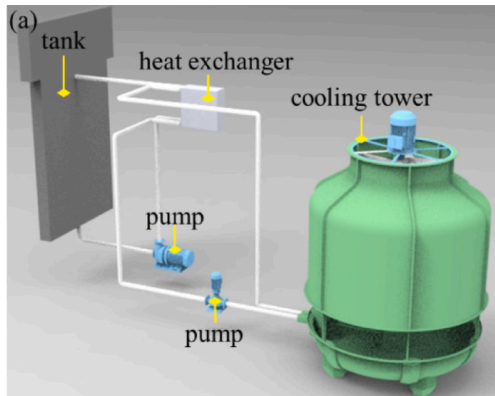


Fig. 4. Model validation: (a) Experimental setup, (b) Result comparison.

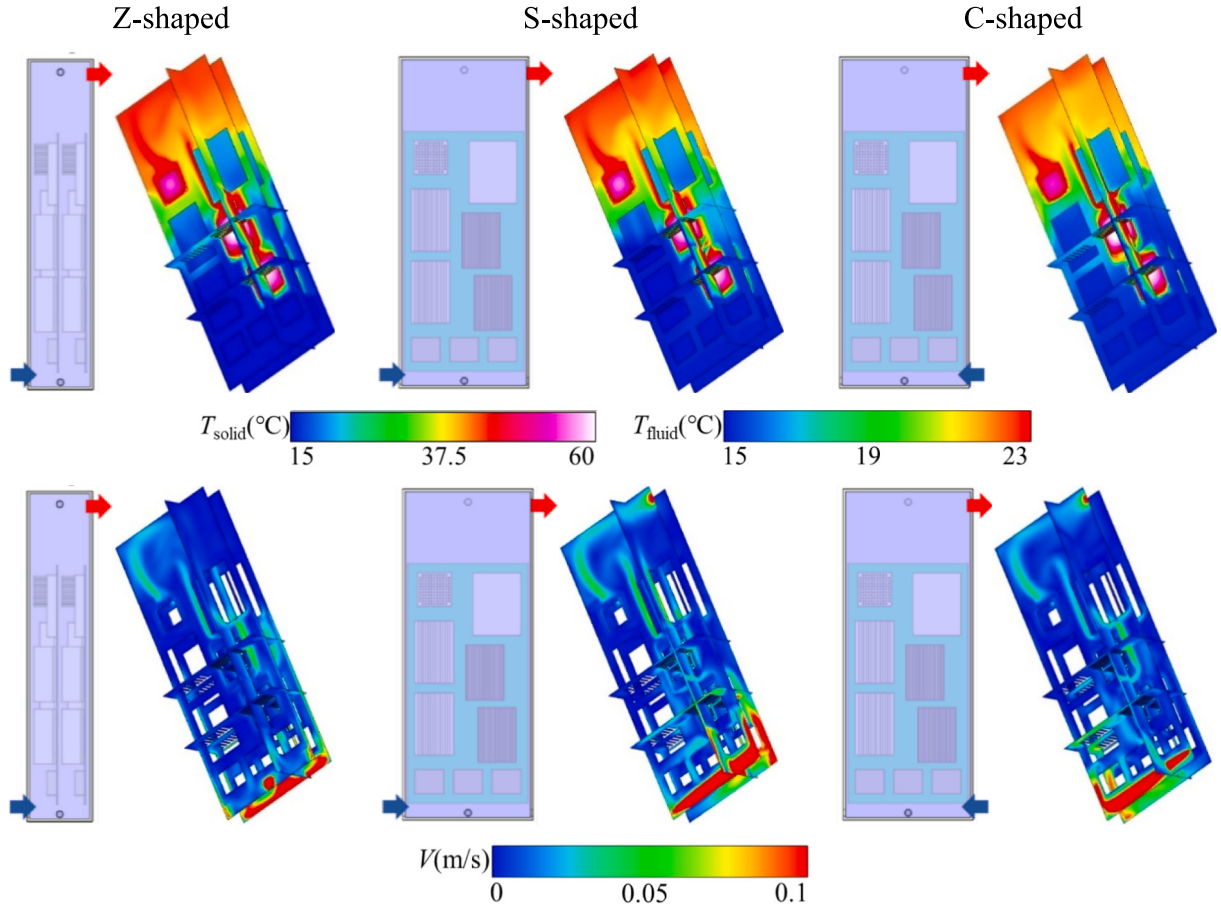


Fig. 5. Temperature and velocity distribution characteristics with different import/export layouts.

3. Results and discussion

3.1. Performance of single-phase ILC tanks with different liquid - cooling structures

To investigate the effects of different liquid - cooling structures on the ILC performance, this paper conducts research on the temperature and velocity distribution characteristics, the flow and heat transfer performance, on the cross - sections where the CPUs are located ($x = 190$ mm, $y = 172.5$ mm and 302.5 mm, $z = 13.5$ mm and 72 mm).

3.1.1. Effect of import/export layout

Fig. 5 shows the temperature and velocity distribution characteristics within the single-phase ILC tank under different import/export layouts.

By observing the temperature contour plot, it is observed that the internal temperature distribution of the single-phase ILC tank is roughly the same under different import/export layouts. Upon closer inspection, it is found that the temperature difference of CPU1 is not significant under different import/export layouts, while there is a certain difference in the temperature of CPU2. This is because CPU1 is located upstream in the flow direction, so the heat transfer capabilities of different coolants are relatively strong. In contrast, CPU2, which is downstream in the flow direction, not only has an increased temperature of the inlet coolant but is also affected by the upstream thermal cascade. Therefore, its performance is more sensitive to the thermophysical properties of the coolant. From the velocity contour plot, it can be seen that the velocity of the coolant entering the cabinet rapidly decreases due to the impact with the tank body, but the impact positions are different under different import/

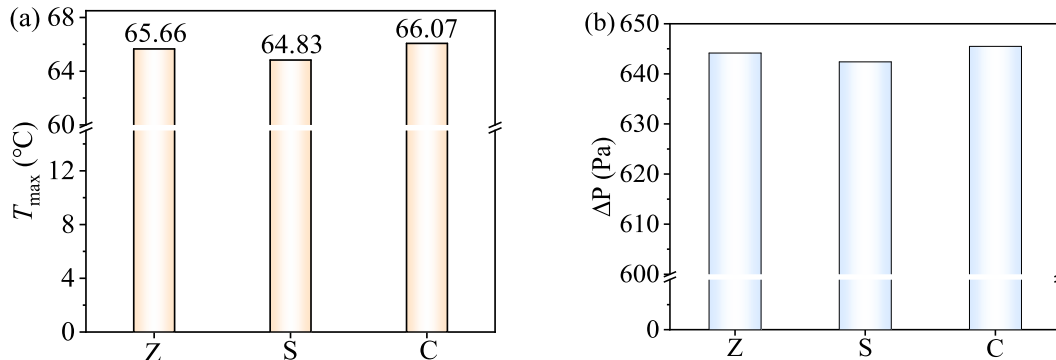


Fig. 6. Convective heat transfer performance with different import/export layouts: (a) Maximum CPU temperature, (b) Pressure drop.

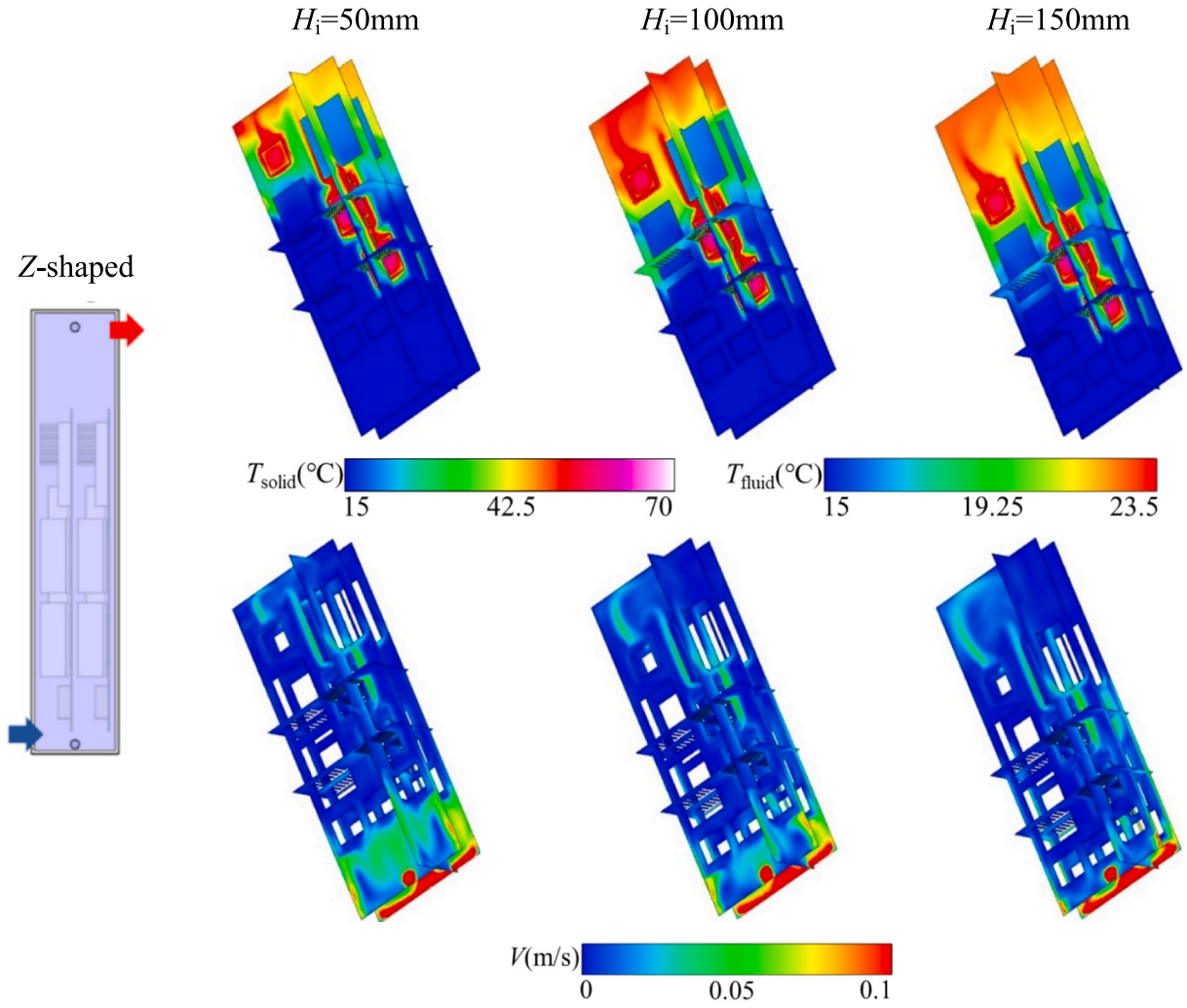


Fig. 7. Temperature and velocity distribution characteristics with different immersion depths.

export layouts. The coolant under the C-shaped liquid cooling structure will impact the side wall of the DIMM, resulting in a relatively large velocity near the DIMM and a relatively small velocity near the CPU. Under the Z-shaped liquid cooling structure, the coolant will impact the rear wall of the tank and then slowly flow back and rise to the vicinity of the CPU. For the S-shaped liquid cooling structure, the coolant directly impacts the side wall of the CPU, making the flow velocity near the CPU relatively high. Therefore, the temperatures of CPU1 and CPU2 under the S-shaped import/export layout are relatively low.

Fig. 6 further presents the maximum CPU temperature (i.e., T_{max}) and pressure drop of the tank (i.e., ΔP) under different working conditions to quantitatively compare the impact of various import/export arrangements on the single-phase ILC performance. It should be noted that a volumetric average power (i.e., a uniform heat source model) is adopted in this study, and the maximum temperature within the CPU generally occurs near the center of its volume. As is observed in Fig. 6(a), compared with the Z-shaped and C-shaped liquid cooling structures, the T_{max} under the S-shaped import/export layout decreases by 1.28% and 1.91%, respectively, which verifies the conjecture mentioned above. In other words, the flow velocities of the coolant near the CPU vary under different import/export layouts, which in turn leads to changes in the convective heat transfer performance of the CPU and its surrounding area. From Fig. 6(b), it can be observed that the import/export layout has little influence on the pressure drop of the single-phase ILC tank. Through quantitative analysis, it is known that the pressure drop under the S-shaped import/export layout is only 3.1 Pa less than that of the

other two liquid cooling structures. Thus, it can be seen that the positions of the import and export have an impact on the flow and heat transfer characteristics within the single-phase ILC tank, but the resulting differences in convective heat transfer performance are not significant. Therefore, in practical engineering, the positions of the import and export of the single-phase ILC tank can be reasonably configured according to the on-site conditions.

3.1.2. Effect of immersion depth

Considering that the price of the fluorinated liquid is too expensive, and the smaller the immersion depth of the simulated MB is, the less fluorinated liquid is required. Given this, it is necessary to research the impact of different immersion depths of the simulated MB on the flow and heat transfer performance within the single-phase ILC tank. Fig. 7 shows the temperature and velocity distribution characteristics within the single-phase ILC tank under different immersion depths. It can be found that the differences in the temperature field and velocity field in the heat source module area under different immersion depths are relatively small, and only the temperature and velocity distribution characteristics at the inlet and outlet areas are different. This phenomenon indicates that the immersion depth has little impact on the thermal management capacity of electronic devices, and only affects the flow and thermal transport characteristics of the coolant area. It can also be inferred that when the coolant flows from the inlet to the bottom of the simulated MB, its flow has been fully developed. At this time, increasing or decreasing the distance between the electronic device and the

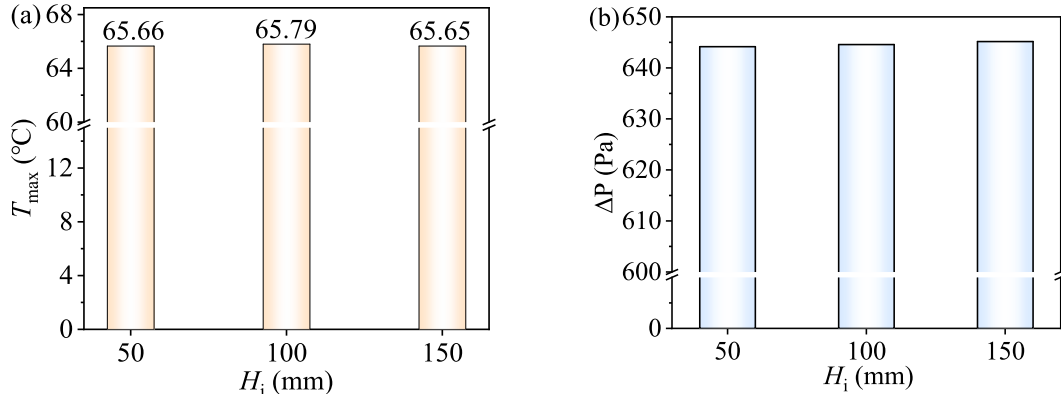


Fig. 8. Convective heat transfer performance with different immersion depths: (a) Maximum CPU temperature, (b) Pressure drop.

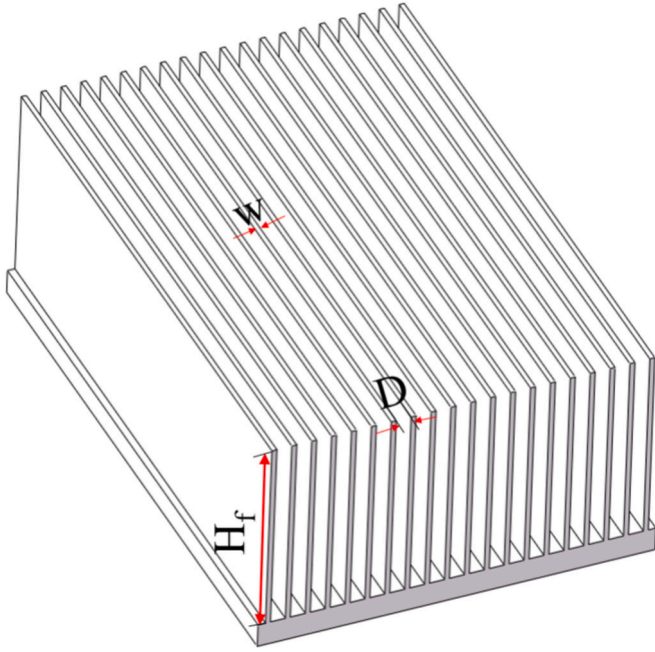


Fig. 9. Schematic diagram of the CPU heat sink.

disturbance source (the coolant inlet) can no longer enhance or weaken the heat dissipation performance of the electronic device.

Fig. 8 further presents the maximum CPU temperature and the pressure drop of the tank under different immersion depths. As can be

seen from Fig. 8(a), there is basically no difference in $T_{\max}$ under different immersion depths, and the maximum deviation is only 0.14 °C. This further shows that changing the immersion depth cannot enhance the heat dissipation performance of the electronic devices. Besides, as is observed in Fig. 8(b), the immersion depth also has little impact on ΔP . Through quantitative evaluation, it is found that the pressure drop of the tank under different immersion depths only differs by 1.01 Pa. In summary, the immersion depth has little influence on the flow and thermal transport characteristics within the single-phase ILC tank. Therefore, in practical engineering, the volume of the single-phase ILC tank should be reduced as much as possible to decrease the consumption of the fluorinated liquid, thereby improving the economic efficiency of the immersion liquid cooling data center.

3.2. Heat sink performance enhancement for single-phase ILC systems

As a component closely related to the CPU, the core part of the simulated MB, the structure of the heat sink is crucial to the CPU's cooling performance. However, the correlation mechanism between the heat sink structure in the current pump-driven single-phase ILC system and the flow and thermal transport characteristics of electronic devices is not yet clear. Meanwhile, the responses of electronic devices to different heat sink structures may vary under different immersion coolants. As shown in Fig. 9, this paper mainly explores the influence of the changes in three parameters on the ILC performance of the heat sink, including the fin pitch (i.e., D), the fin height (i.e., H_f), and the fin thickness (i.e., w). In this paper, the values of D are set to 1 mm, 2 mm, 3 mm, and 4 mm; H_f is set to 19 mm, 26 mm, 33 mm, 40 mm, and 47 mm; and w is set to 1 mm, 2 mm, 3 mm, and 4 mm.

The Nusselt number quantifies the ratio of convective to conductive heat transfer intensity. Modifications to the geometric structure of the

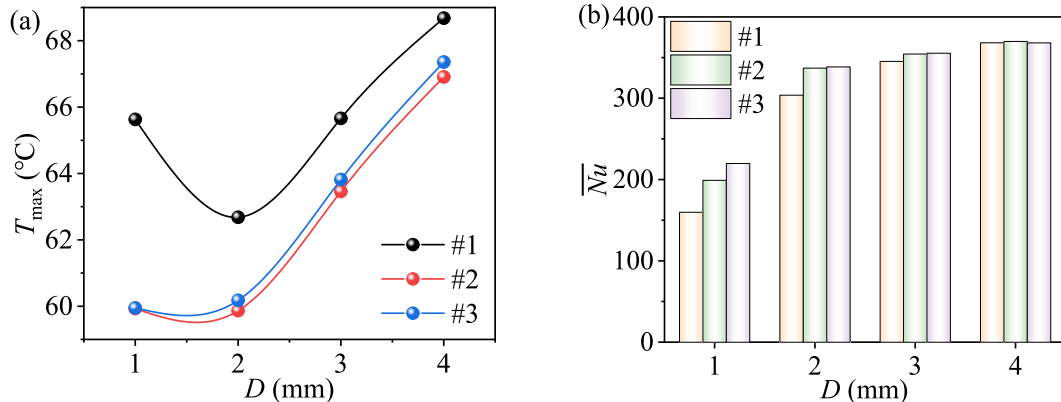


Fig. 10. Effect of fin pitch on the ILC performance: (a) Maximum CPU temperature, (b) Average Nusselt number.

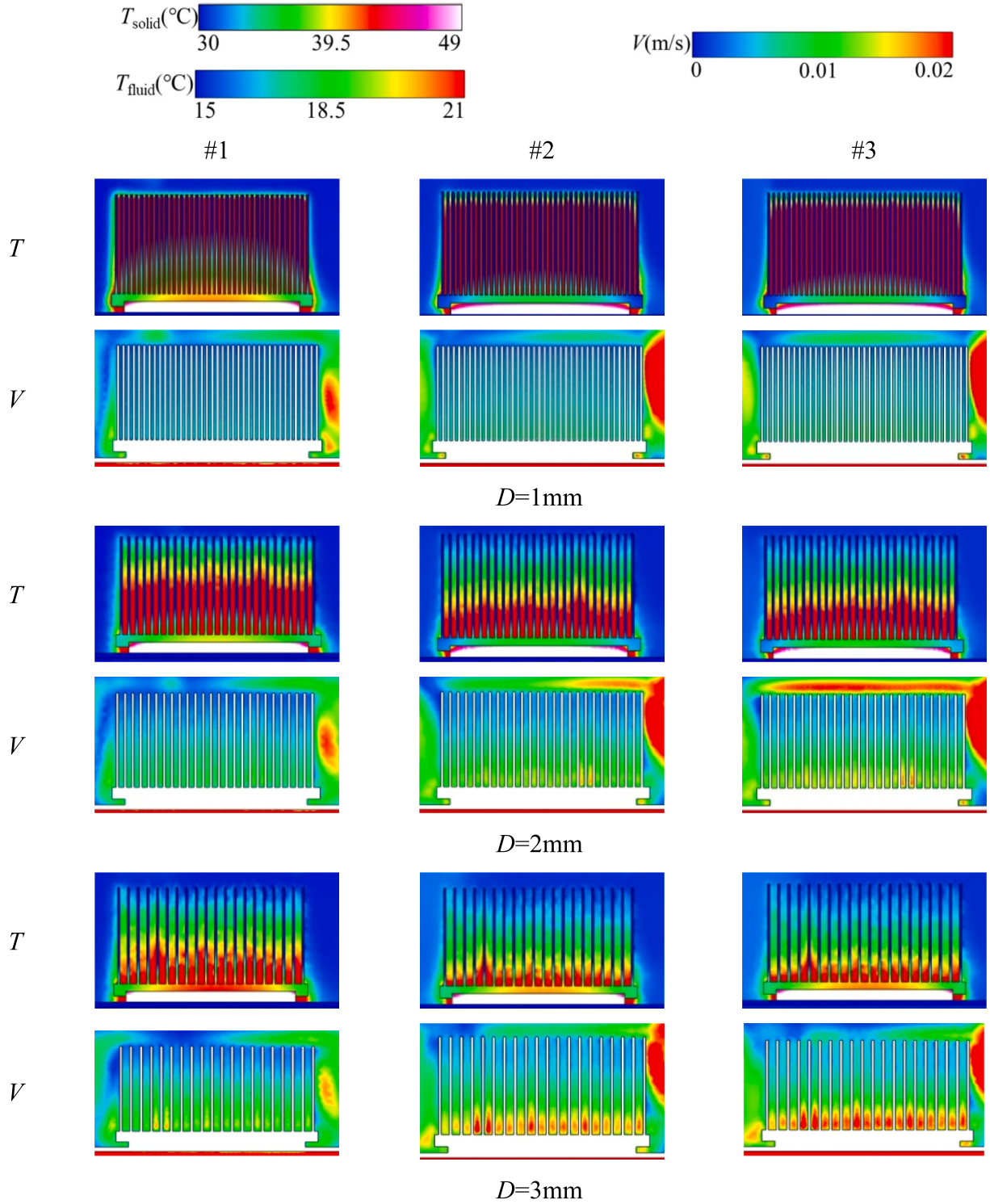


Fig. 11. Effect of fin pitch on the temperature and velocity distribution characteristics.

heatsink simultaneously influence both its heat transfer and flow characteristics. As a key indicator, the average Nusselt number (i.e., $\overline{Nu}$) effectively reflects the convective heat transfer performance of the heatsink.

$$\overline{Nu} = \frac{Pl}{A\Delta T\lambda} \quad (7)$$

where P is the power of a simulated chip, l and A denote the characteristic length and surface area of heat sinks, respectively, and ΔT

denotes the average temperature variation between the heat sink and the coolant.

3.2.1. Effect of fin pitch

To quantitatively evaluate the impact of fin pitch on the single-phase ILC characteristics of the heat sink, Fig. 10 presents the maximum CPU temperature and the average Nusselt number under various fin pitches. For a fair comparison, the height of the heat sink is fixed at 40 mm, and the width is set to 1 mm in all cases. From Fig. 10(a), it can be observed

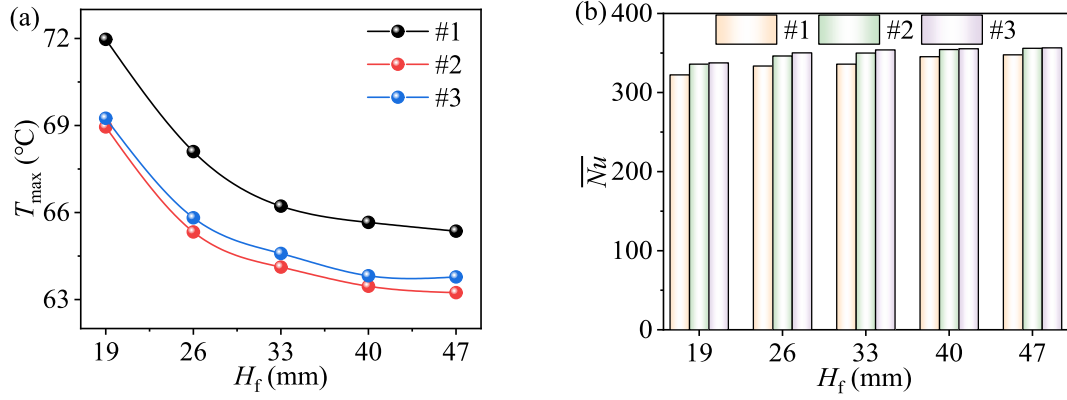


Fig. 12. Effect of fin height on the ILC performance: (a) Maximum CPU temperature, (b) Average Nusselt number.

that the $T_{\max}$ is the highest under coolant #1, while the $T_{\max}$ under coolant #2 is lower than that under coolant #3. Additionally, when the fin pitch ranges between 2 and 4 mm, the $T_{\max}$ under all three coolants increases with larger fin pitch. Quantitative analysis reveals that as the fin pitch increases from 2 mm to 4 mm, the $T_{\max}$ rises by 9.57%, 11.76%, and 11.93% for coolants #1, #2, and #3, respectively. This phenomenon stems from the competing mechanisms between enhanced convective heat transfer due to increased fin pitch and the deterioration of heat dissipation caused by reduced heat transfer area. Furthermore, the sensitivity of temperature response varies among the coolants. Coolant #1, with the highest dynamic viscosity, exhibits a more gradual temperature response to changes in fin pitch, while the other two coolants show greater sensitivity. This is because, for high-viscosity coolants, the reduction in flow resistance significantly enhances convective heat transfer when fin pitch increases, mitigating the negative effects of reduced heat transfer area. However, when the fin pitch is reduced to 1 mm, the $T_{\max}$ under coolant #1 increases again by 4.71%, while the $T_{\max}$ under coolants #2 and #3 remain almost unchanged. This indicates that for high-viscosity coolants, the sharp increase in flow resistance under extreme fin pitch significantly weakens convective heat transfer, leading to a temperature rise. In contrast, for low-viscosity coolants, the competing effects of convective heat transfer and heat transfer area balance out, resulting in negligible temperature changes. A closer examination of Fig. 10(b) reveals that when the fin pitch ranges between 2 and 4 mm, the $\overline{Nu}$ under all three coolants increases with larger fin pitch. Specifically, $\overline{Nu}$ increases by 21.18% for coolant #1, which has the largest viscosity, while it only increases by 8.74% for coolant #3, which has the smallest viscosity. When the fin pitch is reduced from 2 mm to 1 mm, $\overline{Nu}$ drops sharply for all three coolants, decreasing by 47.38%, 40.94%, and 35.07% for coolants #1, #2, and #3, respectively. This further demonstrates that the increased flow resistance caused by reduced fin pitch severely deteriorates convective heat transfer performance. In summary, the temperature response of the CPU to changes in fin pitch varies depending on the coolant used. Excessively reducing fin pitch to increase the heat transfer area may lead to a deterioration in heat sink performance.

Fig. 11 displays the temperature and velocity distributions of the CPU and its heat sink cross-section ($y = 172.5$ mm) under different fin pitches. As the fin pitch increases, the temperatures of the CPU and the heat sink base under different coolants significantly rise. This is because when the fin pitch increases, the reduction in heat transfer performance caused by the decrease in heat sink area outweighs the enhancement in convective heat transfer performance due to reduced flow resistance. Moreover, the coolant temperature in the fin gaps is significantly lower under larger fin pitches compared to smaller fin pitches. This is because as the fin pitch decreases, the coolant velocity in the fin gaps continuously decreases. When the fin pitch is 1 mm, the coolant velocity in the fin gaps drops sharply, almost forming a stagnant flow zone, where heat

transfer relies solely on thermal conduction. Under these conditions, the increase in heat transfer area cannot compensate for the deterioration in convective heat transfer caused by the reduced fin pitch, leading to a sharp rise in coolant temperature. Close observation reveals that the temperatures of the CPU and the heat sink base under coolants #2 and #3 are significantly lower than those under coolant #1, and the average coolant temperature in the fin gaps is also lower. This is because coolant #1 has a higher dynamic viscosity, resulting in a thicker thermal boundary layer on the fin surfaces and an overall lower flow velocity, making it difficult to transfer heat away and causing heat accumulation. Consequently, the immersion cooling performance under coolant #1 is the poorest. This demonstrates that the dynamic viscosity of the coolant has a non-negligible impact on the performance of heat sinks with different fin pitches.

3.2.2. Effect of fin height

Fig. 12 illustrates the variations in the maximum CPU temperature and the average Nusselt number under different fin heights. In this section, the other structural parameters of the heat sink are set as follows: $D = 3$ mm and $w = 1$ mm. As shown in Fig. 12(a), when the fin height ranges between 19 and 47 mm, the $T_{\max}$ under all three coolants decreases with increasing fin height, but the rate of decrease gradually diminishes. Quantitative analysis reveals that as the fin height increases from 19 mm to 47 mm, the $T_{\max}$ decreases by 9.18%, 8.28%, and 7.9% for coolants #1, #2, and #3, respectively. This phenomenon occurs because a reduction in fin height not only increases bypass flow and reduces the coolant flow rate through heat sinks but also significantly decreases the heat transfer area, weakening the convective heat transfer intensity between the coolant and heat sinks, thereby leading to a continuous rise in $T_{\max}$. However, when the fin height increases from 40 mm to 47 mm, the $T_{\max}$ under coolants #1, #2, and #3 decreases by only 0.3 °C, 0.22 °C, and 0.04 °C, respectively. This indicates the existence of a critical fin height beyond which further increases do not significantly enhance the single-phase ILC performance of heat sinks. Additionally, the largest variation in $T_{\max}$ occurs under coolant #1, primarily due to its highest dynamic viscosity, which results in the greatest flow resistance through the heat sink and diverts more coolant to the bypass path, thereby weakening the heat transfer effectiveness of the heat sink. Moreover, because coolant #1 has a relatively poor thermal performance, it is more sensitive to the deterioration in heat transfer caused by a reduction in heat transfer area under the same conditions. From Fig. 12 (b), when the fin height ranges between 19 and 47 mm, the $\overline{Nu}$ under all three coolants increases with fin height, with $\overline{Nu}$ rising by 7.86%, 5.96%, and 5.63% for coolants #1, #2, and #3, respectively. However, when the fin height is further increased to 47 mm, the corresponding values for the three coolants increase by only 0.66%, 0.45%, and 0.33%, respectively.

Fig. 13 illustrates the temperature and velocity distribution

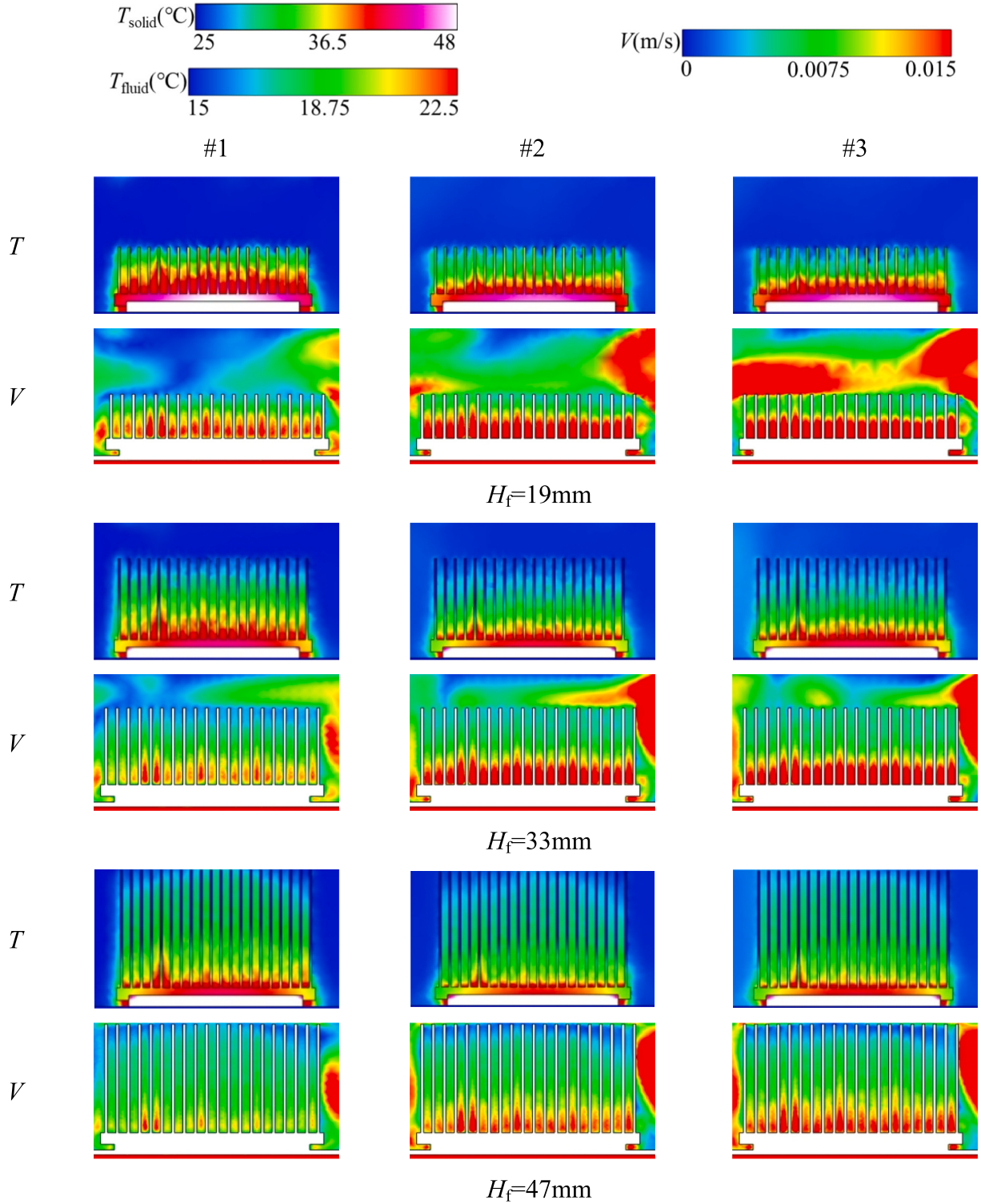


Fig. 13. Effect of fin height on the temperature and velocity distribution characteristics.

characteristics under different fin heights. When the fin height increases from 19 mm to 33 mm, the temperature of the CPU and the heat sink base decreases significantly. However, as the height further increases to 47 mm, the temperature change tends to stabilize, indicating that there is a limit to the improvement in heat transfer performance with increasing fin height. Observing the temperature distribution of the coolant between the fins reveals that the high-temperature region remains almost unchanged as the fin height increases. This suggests that the increased heat transfer area due to higher fins allows the heat sink to

transfer more heat to the coolant, while the enhanced heat transfer effect enables the coolant to dissipate the heat more efficiently. Additionally, the temperatures of the CPU and the heat sink base under coolants #2 and #3 are significantly lower than those under coolant #1, especially at a fin height of 19 mm. This is because the high dynamic viscosity of coolant #1 makes it more sensitive to the bypass effect, resulting in poorer immersion cooling performance. The velocity distribution shows that as the fin height increases from 19 mm to 47 mm, the local velocity between the fins slightly decreases. This is due to the increased cross-

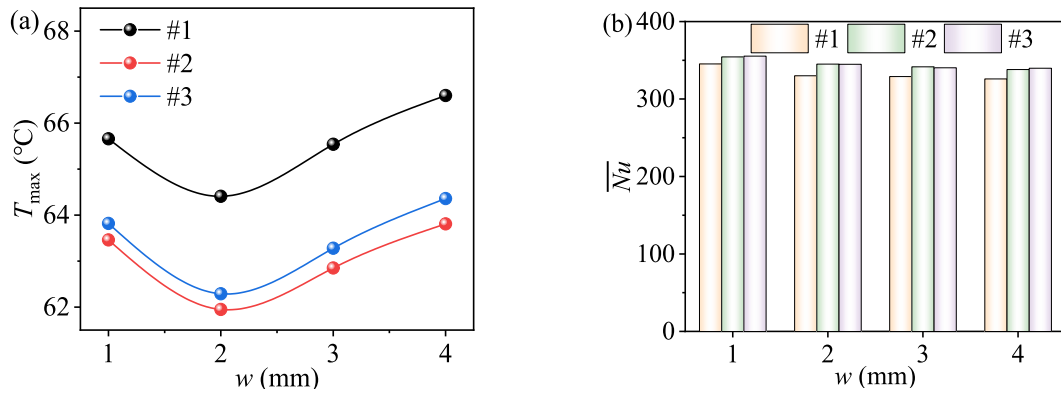


Fig. 14. Effect of fin thickness on the ILC performance: (a) Maximum CPU temperature, (b) Average Nusselt number.

sectional area between the fins, which reduces the flow velocity, while the reduced bypass effect increases the coolant flow rate through the heat sink. The velocities of coolants #2 and #3 are similar and higher than that of coolant #1, and they increase with fin height. This is because the lower dynamic viscosity of coolants results in smaller flow resistance, making the impact of increased fin cross-sectional area on velocity less pronounced. When the fin height increases from 40 mm to 47 mm, the change in coolant velocity becomes most noticeable, indicating that the dominant effect of increased fin cross-sectional area reduces the flow velocity, while the reduced bypass effect no longer significantly increases the flow rate. This further demonstrates that increasing fin height cannot indefinitely improve the convective heat transfer performance between the coolant and heat sinks.

3.2.3. Effect of fin thickness

To quantitatively evaluate the impact of fin thickness on the single-phase ILC performance of heat sinks, Fig. 14 plots the variations in the maximum CPU temperature and the average Nusselt number under different fin thicknesses. In the cases studied, the fin pitch of the heat sink is fixed at 3 mm, and the fin height is set to 40 mm. As is observed in Fig. 14(a), when the fin thickness varies within the range of 1–4 mm, the $T_{\max}$ under all three coolants exhibits a trend of first decreasing and then increasing, with relatively small overall variations. Quantitative analysis reveals that when the fin thickness increases from 1 mm to 2 mm, the $T_{\max}$ decreases by 1.25 °C, 1.51 °C, and 1.53 °C for coolants #1, #2, and #3, respectively. This phenomenon can be attributed to the competing mechanisms between enhanced heat transfer due to increased contact area between the fins and the base and the deterioration in heat transfer caused by reduced heat exchange area. However, when the fin thickness further increases to 4 mm, the $T_{\max}$ rises by 2.19 °C, 1.86 °C, and 2.07 °C for coolants #1, #2, and #3, respectively. This indicates that excessive fin thickness significantly reduces the heat exchange area, weakening the convective heat transfer performance between the coolant and heat sinks, and the enhanced heat transfer effect due to the increased contact area between the fins and the base cannot compensate for the weakened convective heat transfer. From Fig. 14(b), as the fin thickness increases from 1 mm to 4 mm, the $\overline{Nu}$ under all three coolants decreases, with reductions of 5.61%, 4.6%, and 4.38% for coolants #1, #2, and #3, respectively. This demonstrates that increasing the contact area between the fins and the base only enhances the heat conduction process from the CPU to the fins but cannot compensate for the weakened convective heat transfer between the coolant and heat sinks. Therefore, the optimal design of fin thickness requires a balance between the contact area and the heat exchange area to maximize cooling performance.

Fig. 15 presents the temperature and velocity distribution cloud maps under different fin thicknesses, revealing the influence mechanism of fin thickness on the performance of immersion liquid-cooled heat sinks. Compared to coolant #1, the temperatures of the CPU and the

heat sink base under coolants #2 and #3 are lower. When the fin thickness increases from 1 mm to 2 mm, the corresponding temperatures decrease significantly, indicating that the increased contact area between the fins and the base enhances heat conduction from the base to the fins. However, when the fin thickness increases from 2 mm to 3 mm, both the base temperature and the coolant temperature rise noticeably, suggesting that the enhanced heat transfer due to the increased contact area cannot compensate for the deterioration in thermal transport performance caused by the reduced heat exchange area. Additionally, the velocity distribution cloud map shows that the flow velocities of coolants #2 and #3 between the fins are significantly higher than that of coolant #1. This is because the high dynamic viscosity of coolant #1 results in poorer fluidity, making it prone to forming a viscous layer on the fin surfaces. As the fin thickness increases from 1 mm to 3 mm, the overall coolant flow velocity rises, as the increased cross-sectional area occupied by the fins narrows the coolant flow channels, thereby increasing the local flow velocity. This phenomenon further demonstrates that the geometric dimensions of the heat sink have a significant impact on the flow characteristics of the coolant.

3.3. Comprehensive evaluation of heat sink performance

The aforementioned research demonstrates that the geometric parameters of heat sinks significantly influence its flow and thermal transport characteristics, providing critical guidance for the design of immersion liquid-cooled heat sinks. Fig. 16 illustrates the impact of changes in heat sink structural parameters on the maximum CPU temperature under different coolants. As shown in the figure, alterations in the heat sink geometric parameters significantly affect the $T_{\max}$. Statistical analysis reveals that when the heat sink structure is varied, the $T_{\max}$ fluctuation ranges for coolants #1, #2, and #3 are 62.68–71.97 °C, 59.87–68.95 °C, and 59.95–69.25 °C, respectively. Although the $T_{\max}$ fluctuation ranges under the three coolants are similar, the $T_{\max}$ under coolant #1 is higher due to its weaker convective heat transfer performance with the heat sink. Further analysis indicates that changes in fin pitch and fin height have a more pronounced impact on $T_{\max}$, while changes in fin thickness have a relatively minor effect. Quantitative analysis shows that when the fin pitch increases from 1 mm to 4 mm, the $T_{\max}$ under coolants #2 and #3 rise by 6.98 °C and 7.41 °C, respectively. Conversely, when the fin height increases from 19 mm to 47 mm, the $T_{\max}$ under coolant #1 decreases by 6.61 °C. This phenomenon indicates that coolants with different thermophysical properties exhibit varying temperature responses to changes in heat sink geometry. Due to its larger viscosity, coolant #1 experiences greater flow resistance through the heat sink, making it more sensitive to the deterioration in convective heat transfer caused by reduced fin pitch. In extreme cases, excessively small fin pitch can lead to severe heat transfer degradation. Additionally, coolant #1 is more sensitive to the bypass effect and the reduction in heat transfer area caused by decreased fin height, which is why its

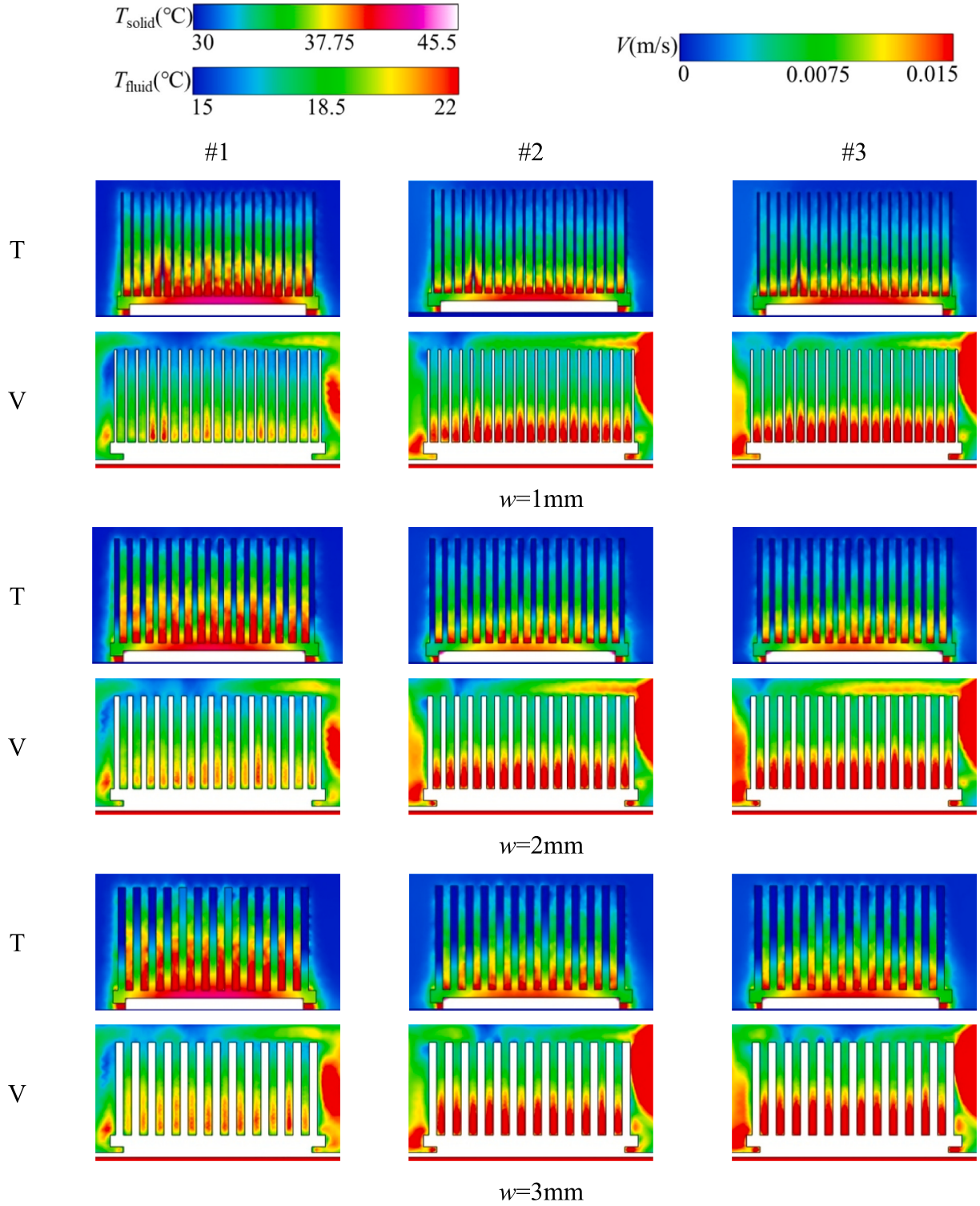


Fig. 15. Effect of fin thickness on the temperature and velocity distribution characteristics.

T_{max} fluctuations primarily occur with changes in fin height. In contrast, coolants #2 and #3, with their lower dynamic viscosity, exhibit T_{max} fluctuations primarily with changes in fin pitch.

In summary, the geometric structure of immersion liquid-cooled heat sinks, particularly fin pitch and fin height, plays a crucial role in the convective heat transfer performance. In practical design, the heat sink structure should be optimized according to the thermophysical properties of the coolant to avoid significant deterioration in heat transfer performance.

4. Conclusions

To investigate the impact of the coolant properties and the liquid cooling structure (including import/export layout, immersion depth, and immersion liquid cooling heat sink) on the single-phase ILC performance, this work established a 3D steady-state model of the flow and thermal transport processes within a single-phase ILC tank. The study explored the effects of the import/export layout of the single-phase ILC tank, the immersion depth of the simulated MB, and the heat sink

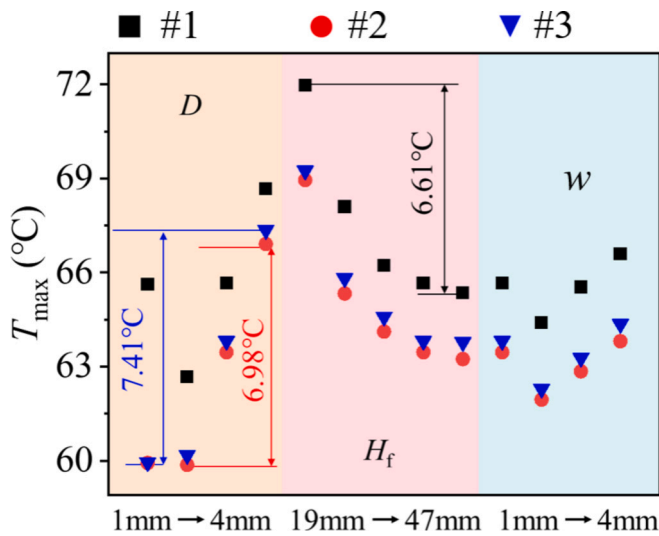


Fig. 16. Comprehensive evaluation of heat sinks for single-phase ILC systems.

structure on the single-phase ILC performance. Furthermore, the intrinsic relationship between the coolant properties and the cooling performance of the single-phase ILC tank under changes in the heat sink structure was clarified, providing theoretical guidance for practical engineering applications. Based on the findings of this study, the main conclusions are as follows.

- (1) Under identical conditions, the import/export layout of the single-phase ILC tank and the immersion depth of the simulated MB have a small impact on the ILC performance. Specifically, the maximum CPU temperature under the S-shaped import/export layout is 1.28% and 1.91% lower than that under the Z-shaped and C-shaped arrangements, respectively. Additionally, the maximum CPU temperature difference and pressure drop difference under different immersion depths are 0.14 °C and 1.01 Pa, respectively.
- (2) The geometric structure of the heat sink has an important influence on the single-phase ILC performance. As the fin pitch decreases, the ILC performance of the heat sink continuously enhances. However, when the fin pitch is lower than 2 mm, the immersion liquid cooling performance will deteriorate. Increasing the fin height is beneficial to the improvement of the immersion liquid cooling performance, but there is a critical fin height (40 mm). When it exceeds this value, the immersion liquid cooling performance is insensitive to the fin height. As the fin thickness increases, the single-phase ILC performance of heat sinks first improves and then deteriorates again, and the immersion liquid cooling performance reaches its best when the fin thickness is 2 mm.
- (3) The influence of the thermal physical properties of the coolant on the single-phase ILC performance varies among different liquid cooling structures. Among them, the dynamic viscosity of the coolant has the most significant impact on the single-phase ILC performance. Especially in liquid cooling structures with different fin pitches, the attenuation of the cooling performance of the coolant with small viscosity is much lower than that of the coolant with large viscosity when the fin pitch is less than 1 mm. However, the sensitivity of the influence of the coolant properties on the single-phase ILC performance to the fin height and fin thickness is relatively not obvious.

This study systematically investigates the influence of heatsink structural parameters and coolant properties on ILC performance, with a focus on revealing the coupling mechanism between fin geometries and

the thermophysical properties of coolants. However, a systematic dimensional analysis to derive more universal heat transfer and flow correlations has not yet been conducted. Given that ILC performance is inherently governed by the coupled effects of three major factors—fluid operating conditions, fin structures, and coolant physical properties—future research will introduce dimensionless numbers such as the Nusselt number (Nu), Prandtl number (Pr), and Reynolds number (Re), along with dimensionless treatments of structural parameters. Based on this, a universally applicable dimensionless empirical correlation will be developed through data correlation fitting. This will enable the generalization of the conclusions obtained under specific conditions in this paper to a broader range of engineering application scenarios.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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